Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Separate English and Telugu Models

Technical Report

Repository	https://github.com/Lohith0077/FastSpeech2-English-Telugu-TTS
Datasets	English: AI4Bharat Svarah | Telugu: SPRINGLab IndicTTS Telugu
Training platform	Kaggle Notebook, NVIDIA T4 GPU
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1. Implementation Approach

Two independent FastSpeech 2 models were trained, one for English and one for Telugu, using a PyTorch FastSpeech 2 implementation as the starting point (configuration layout `config/LJSpeech/{preprocess,model,train}.yaml`). [TO FILL: confirm the exact upstream repository name and link.] The two languages share the same architecture and training recipe and differ in their text front-end, symbol inventory, alignment resources and datasets. Waveforms are produced with a pretrained Universal HiFi-GAN vocoder in both cases.

1.1 Architecture components used from the reference implementation

Component	Description	Setting
Phoneme / symbol embedding	Maps input tokens to hidden vectors; positional encoding added	Hidden dim 256
Encoder	Feed-forward Transformer (FFT) blocks: multi-head self-attention + 1D convolution	4 layers, 2 heads
Variance adaptor	Duration, pitch and energy predictors (conv + layer norm + dropout). Duration is used by the length regulator to expand the encoder output to mel-frame length. Pitch and energy are predicted, quantised into bins and added as embeddings	Filter size 256; 256 pitch bins; 256 energy bins; phoneme-level pitch/energy; normalisation on
Mel decoder	FFT blocks producing the mel-spectrogram, followed by a convolutional postnet	6 layers, 2 heads, 80 mel channels
Vocoder	Universal HiFi-GAN (pretrained, not retrained)	22,050 Hz
Loss	Mel (L1) + postnet mel (L1) + duration (MSE, log domain) + pitch (MSE) + energy (MSE)	Total = sum of components

1.2 Modifications for each language

Aspect	English	Telugu
Input tokens	ARPAbet phonemes	Telugu Unicode graphemes (characters)
Text front-end	`g2p_en` converts words to phonemes at synthesis time; lowercase normalisation	No G2P: characters are used directly as tokens; Unicode NFC normalisation
Symbol inventory	Standard ARPAbet set	Telugu letters (vowels, consonants, vowel signs, virama, anusvara etc.) added to the model's symbol list
Aspect	**English**	**Telugu**

Forced alignment	Montreal Forced Aligner, pretrained `english_us_arpa` acoustic model and dictionary	MFA with a custom Telugu grapheme dictionary and an acoustic model trained on the Telugu subset
Pitch / energy / duration targets	Extracted from MFA alignments (TextGrids)	Extracted from MFA alignments (TextGrids)
Vocoder	Universal HiFi-GAN	Universal HiFi-GAN

Treating Telugu graphemes as tokens is practical because Telugu orthography is largely phonemic, but conjuncts (consonant + virama + consonant) are represented as sequences of separate characters, so the model must learn their combined acoustics. [TO FILL: list the specific files you modified, e.g. text/symbols.py, text/__init__.py, preprocessor, config files, inference script.]

2. Datasets, Subsampling and Preprocessing

Detail	English	Telugu
Dataset	AI4Bharat Svarah	SPRINGLab IndicTTS Telugu
Hugging Face ID	ai4bharat/Svarah	SPRINGLab/IndicTTS_Telugu
Source split used	test	train
Training set	3,477 utterances, 5.002 h	2,188 utterances, 5.000 h
Validation set	674 utterances, 1.001 h	418 utterances, 1.001 h
Total	4,151 utterances, 6.003 h	2,606 utterances, 6.001 h

Subsampling strategy (both languages). Utterance indices were shuffled with random seed 42. Utterances were then drawn until about 5 hours (training) and about 1 hour (validation) had been collected. The two sets were built as separate, non-overlapping subsets of utterances. Very short and very long utterances were removed during dataset preparation. [TO FILL: the duration thresholds used.]

Preprocessing. Audio was converted to mono and resampled to 22,050 Hz. Mel-spectrograms were computed with FFT size 1,024, window 1,024, hop 256 and 80 mel channels. Text was normalised as described in Section 1.2, forced alignment was run with MFA to obtain phoneme/grapheme durations, and phoneme-level pitch and energy were extracted and normalised (statistics saved by the preprocessing step). Because Svarah is accented, conversational-style English speech from many speakers rather than a single studio voice, the English model has to cope with more speaker and recording variability than the Telugu model. [TO FILL: whether speaker embeddings were used, and how many speakers each dataset has.]

3. Training Configuration

Parameter	Value (both languages)
Total steps	20,000
Batch size / grad. accumulation	8 / 2 (effective batch size 16)
Checkpoints	Every 5,000 steps (5000, 10000, 15000, 20000); final model = 20000.pth.tar
Checkpoint size	English 398.74 MB, Telugu 398.98 MB each
Validation interval	1,000 or 2,000 steps depending on the language config [TO FILL: check the YAML and state one value per language]
Hardware	NVIDIA T4 on Kaggle
Training time	Telugu: 1 h 55 min 7 s. English: [TO FILL: from Kaggle log]

With an effective batch size of 16, 20,000 steps correspond to 320,000 utterance presentations, which is about 92 passes over the English training set (3,477 utterances) and about 146 passes over the Telugu training set (2,188 utterances). The schedule

is much shorter than the long schedules typically used for this model on large single-speaker corpora, so the models are likely under-trained in terms of steps, while the small datasets are seen many times.

4. Training and Validation Loss Analysis

Final recorded values at step 20,000:

Loss component	English	Telugu
Total training loss	1.8490	1.5402
Mel loss	0.6939	0.6561
Postnet mel loss	0.6940	0.6553
Pitch loss	0.2381	0.0913
Energy loss	0.1526	0.0951
Duration loss	0.0704	0.0424
Total validation loss	**4.0699**	**4.0750**

Figures: Results/English/training_loss.png, validation_loss.png and Results/Telugu/training_loss.png, validation_loss.png.

Observations from the recorded final values.

1. **Telugu trains to a lower loss than English.** The gap is largest in the variance terms: pitch loss is 0.091 vs 0.238 and energy loss 0.095 vs 0.153. Duration loss is also lower (0.042 vs 0.070). Mel losses are close (0.69 vs 0.66). A plausible reason is that the IndicTTS Telugu data is studio-recorded and speaker-consistent, so pitch and energy are easier to predict, while Svarah has varied speakers, accents and recording conditions.

2. **There is a large generalisation gap in both models.** Validation total loss is about 4.07 for both languages, more than twice the final training loss (1.85 and 1.54). The validation losses of the two languages are almost identical even though their training losses differ, which suggests that Telugu fits its training data more tightly without generalising better. This is consistent with Telugu having seen its training set about 146 times against about 92 for English. Two cautions: the training value is typically the logged loss of a recent batch, while the validation value is an average over the whole validation set, so the two are not perfectly comparable; and the per-component validation losses should be checked to see which term drives the gap (variance predictors usually generalise worst on small data).

3. **Mel and postnet losses are nearly identical** in both languages (0.6939 vs 0.6940 and 0.6561 vs 0.6553), i.e. the postnet adds essentially no refinement at this stage of training, another sign that the models are short of training.

4. Overfitting cannot be confirmed or ruled out without the full validation curve. If validation loss wasstill decreasing at step 20,000, the gap reflects under-training plus small-data variance; if it flattened or rose earlier, it is overfitting and an earlier checkpoint (e.g. 10,000 or 15,000) may sound better. [TO FILL: state which applies from your curves.]

5. Generated Speech: Test Sentences and Observations

Speech was generated with the final checkpoint (20000.pth.tar) of the matching language model for the sentences specified in the assignment, preserved exactly as given.

ID	Sentence (as specified)	What this sentence tests
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Very long out-of-vocabulary word; G2P and long-sequence duration
E2a	The orchestra's conduct during the ceremony was praised by the audience.	Homograph 'conduct' as a noun (stress on first syllable)
ID	**Sentence (as specified)**	**What this sentence tests**
E2b	The researchers will conduct an experiment to evaluate the proposed method.	Homograph 'conduct' as a verb (stress on second syllable)

E3	The rhyth m of the orchestra changed unexpectedly.	Broken word 'rhyth m' (stray space) and the rare word 'rhythm'
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Time, date and number normalisation; OOV token 'andaroo'
T1	Long technical pharmacology sentence (new drugs' effect analysed)	Loanword-like long word, conjunct consonants
T2a	Sentence about a strong grip on a rope (pattu = grip)	Homograph pattu
T2b	Sentence about wearing a silk saree to a function (pattu = silk)	Homograph pattu in a different sense
T3	Sentence about a tiger walking silently in the forest	Geminate / conjunct in nisshabdamgaa
T4	Sentence with date, time and a request to be ready	Telugu numbers, date and time normalisation

The 'tests' column describes what each sentence is designed to probe and what to listen for; it is not a result.

5.1 Example waveforms and spectrograms

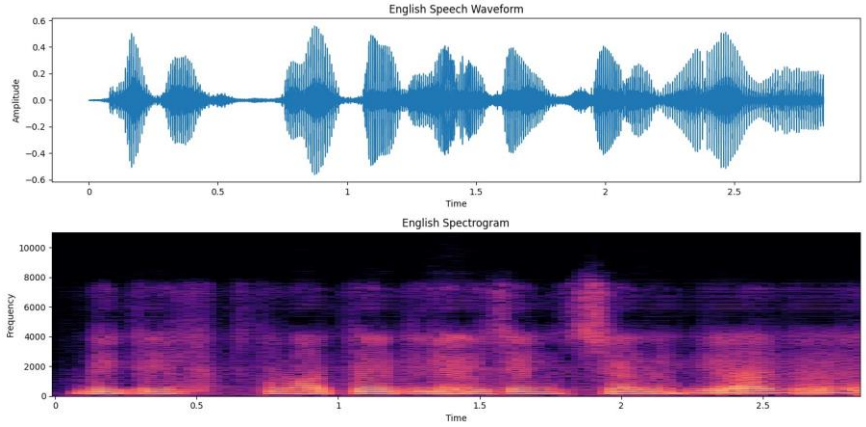


Figure 1. Waveform and spectrogram of a generated English utterance (about 2.8 s).

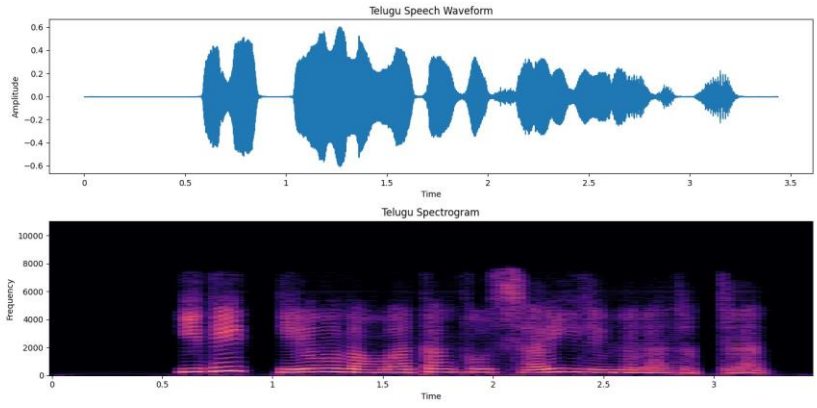


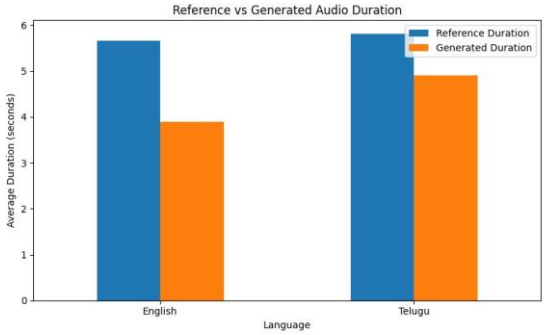
Figure 2. Waveform and spectrogram of a generated Telugu utterance (about 3.4 s).

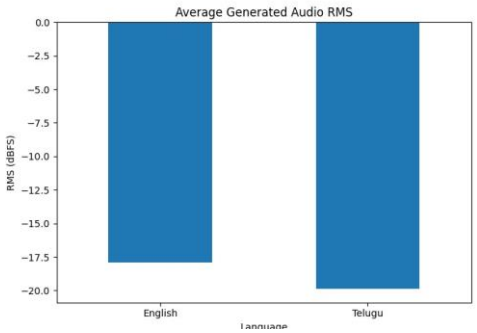
Reading the figures. The English utterance starts almost immediately (speech begins at about 0.1 s), has one clear pause near 0.55 to 0.75 s, and then runs continuously with visible harmonic structure in the spectrogram. The Telugu utterance has about 0.55 s of leading silence, short gaps near 0.9 to 1.0 s and 3.0 s, and trailing silence after about 3.3 s. This leading and trailing silence is consistent with the larger near-silence proportion reported for Telugu in Section 5.2. In both spectrograms the energy is concentrated below about 8 kHz and is largely absent above it, so the output has a limited high-frequency bandwidth, which can make the speech sound slightly dull. Harmonics are clear in the voiced regions of both utterances. [TO FILL: state which sentence each figure shows, and replace them with plots for the assignment test sentences if you can.]

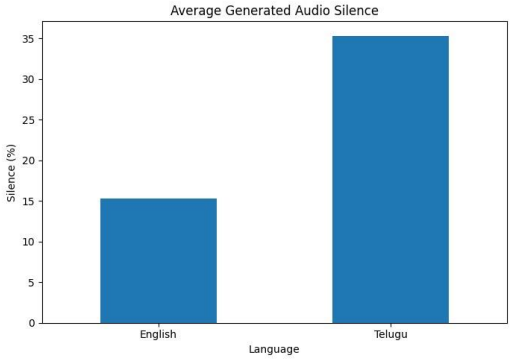
5.2 Aggregate audio measurements (earlier evaluation set)

Metric	English	Telugu
Reference average duration	5.666 s	5.816 s
Generated average duration	3.892 s	4.906 s
Average RMS level	-17.920 dB	-19.898 dB
Near-silence proportion	15.292%	35.352%
Clipping proportion	0%	0%

These numbers come from an earlier evaluation on a different sentence set, so re-compute them on the assignment's test sentences if you quote them as results. They do show two things worth discussing: generated speech is shorter than the reference average (the test sentences differ from references, so this is not directly a speaking-rate measurement), and Telugu output contains a much larger share of near-silence (about 35% vs about 15%), which may indicate long pauses or leading and trailing silence predicted by the duration model. No clipping was found in either language. Overall listening impression recorded earlier: English generally natural with some unclear words; Telugu generally natural with minor instability.







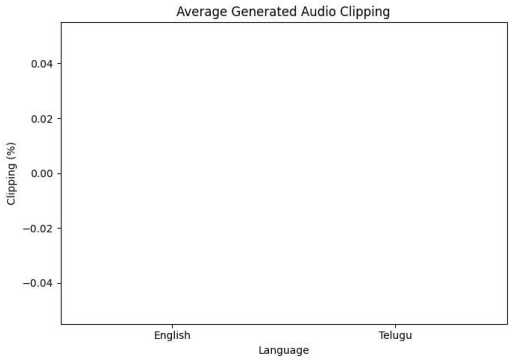


Figure 3. Top: average reference vs generated duration.Figure 4. Top: average near-silence proportion. Bottom: Bottom: average RMS of generated audio (dBFS). average clipping proportion (0% for both, so bars are empty).

Reading Figures 3 and 4. Generated English audio is about 31% shorter than the English reference average (3.89 s vs 5.67 s), while Telugu is about 16% shorter (4.91 s vs 5.82 s); because the generated sentences differ from the reference sentences, this reflects sentence length as much as speaking rate. English is louder on average (-17.9 dBFS vs -19.9 dBFS, about 2 dB). Telugu has more than twice the near-silence of English (about 35% vs about 15%). The clipping panel is empty because the value is exactly 0% for both languages; its axis only spans -0.04 to 0.04, which can look misleading, so you may prefer to state this in the text and drop the panel.

6. Variance Adaptor Analysis (Duration, Pitch, Energy)

For each test sentence, plots of predicted phoneme duration, pitch (F0) and energy are stored under Results/<Language>/variance_adaptor_plots/{duration,pitch,energy}/.

Important note on how the plots were produced. The repository README states that the existing duration, pitch and energy plots are audio-derived analyses (pitch tracked with YIN, energy as RMS, duration from the generated waveform). They are measurements of the synthesised speech, not the outputs of the model's own duration, pitch and energy predictors. The assignment asks for the variance adaptor's predictions, which in a FastSpeech 2 implementation are returned by the model's forward pass during inference. If you can still regenerate the plots from those predictor outputs, do so and delete this

paragraph; otherwise keep it as an honest limitation and interpret the plots as 'what the synthesised audio realises' rather than 'what the predictors output'.

6.1 What to look for in each plot

Plot	What to describe
Duration	Per-phoneme (English) or per-character (Telugu) frame counts. Longer values on stressed vowels, phrase-final syllables and conjuncts; very short values for function words; any abnormally long or zero-length tokens (a sign of alignment or text-frontend failure).
Pitch (F0)	Overall contour shape: declination across the sentence, final fall for statements, local rises on stressed syllables. Compare E2a vs E2b (does stress move?) and T2a vs T2b. Flat or jittery F0 indicates weak prosody modelling.
Energy	Peaks on stressed vowels and loud segments, dips at word boundaries and pauses; whether energy tracks the duration pattern.

7. Comparative Observations, Limitations and Improvements

Supported by the recorded results

- Telugu reached a lower training loss, especially in pitch, energy and duration, but validation loss wasessentially identical (4.07), so there is no evidence that Telugu generalises better.
- Both models show a validation loss more than twice the training loss, indicating a generalisation gapwith about 5 hours of data and about 92 to 146 passes over it.
- Telugu output had a much larger near-silence proportion (about 35% vs about 15%).

Structural differences that explain likely behaviour (to be confirmed against your listening results)

- **Text front-end.** English relies on `g2p_en`, which is dictionary and neural based. It handles common words well but gives one pronunciation per spelling, so the homograph 'conduct' (E2a/E2b), the long word (E1), the broken token 'rhyth m' (E3) and numbers/dates/'andaroo' (E4) are likely failure points. Telugu uses raw graphemes, which avoids G2P errors but gives no explicit handling of numerals, dates and times (T4) and requires the model to learn conjunct pronunciation (T1, T3). The homograph pattu (T2a/T2b) cannot be disambiguated from characters alone.
- **Data.** Svarah consists of many accented speakers and conditions, so English has more variability to model from the same 5 hours, whereas the studio-recorded IndicTTS data is cleaner.
- **Alignment quality.** English uses a pretrained MFA model; the Telugu aligner was trained on the Telugu subset only, so its alignments may be less accurate, which directly affects duration targets and therefore prosody.

Possible improvements

1. Train longer (and monitor validation to pick the best checkpoint), add early stopping or strongerregularisation, and use more data.
2. Add a text-normalisation stage (numbers, dates, times, abbreviations) for both languages, and aTelugu G2P or a phoneme-level lexicon to handle conjuncts and loanwords.
3. Use context-aware disambiguation for homographs (part-of-speech tagging for 'conduct'; context ora lexicon for pattu).
4. Fix input hygiene (collapse stray whitespace) and handle OOV words with a better fallback G2P.
5. Improve Telugu alignments (more data for the aligner, or a multilingual or Indic pretrained aligner),and trim leading and trailing silence in training data to reduce near-silent output.
6. Fine-tune or retrain the vocoder on each language's speech for better naturalness.

8. Reproducibility

Environment. Kaggle Notebook with an NVIDIA T4 GPU. Dependencies are listed in requirements.txt.
[TO FILL: Python and PyTorch versions, MFA version, and the main libraries with versions.]

Configuration files. English/config.yaml and Telugu/config.yaml point to `config/LJSpeech/preprocess.yaml`, `model.yaml` and `train.yaml` inside each language folder.

Commands. Replace the script names with the ones in your repository if they differ. The steps follow the standard FastSpeech 2 workflow; verify each flag against your scripts before submitting.

```
# 0. Setup
pip install -r requirements.txt

# 1. Preprocess (run inside English/ or Telugu/)
#    prepares wavs + transcripts, runs MFA alignment, extracts mel/pitch/energy/duration python
preprocess.py config/LJSpeech/preprocess.yaml

# 2. Train (20,000 steps, batch 8, grad-accum 2) python
train.py -p config/LJSpeech/preprocess.yaml \
               -m config/LJSpeech/model.yaml \
               -t config/LJSpeech/train.yaml

# 3. Inference with the final checkpoint python
inference.py --restore_step 20000 --mode single \
               -p config/LJSpeech/preprocess.yaml \
               -m config/LJSpeech/model.yaml \
               -t config/LJSpeech/train.yaml \
               --text "<sentence>"
```

Checkpoints. English and Telugu checkpoints (5000, 10000, 15000, 20000 steps; about 399 MB each) are too large for a normal GitHub push and are hosted externally. [TO FILL: add the public download links here and in README.md, and confirm they are accessible without login.]

Seed and splits. Utterance shuffling used seed 42. [TO FILL: add the split manifests (train/val file lists) to the repository so the subsets can be reproduced exactly.]